

Three-Tier Epistemic Architecture

Tier 3 — Computational

Currency: prediction errors ϵ , precision Π ,
variational F, d', AUC
Evidence: single-trial EEG, model fit, BIC/AIC

Tier 2 — Information-Theoretic

Currency: bits, nats, mutual information,
channel capacity, Φ
Evidence: transfer entropy, bias-corrected MI, Φ

Tier 1 — Thermodynamic

Currency: joules, ATP, watts, entropy
Evidence: ^{31}P -MRS, two-photon imaging,
BOLD-calibrated fMRI (proxy)

Valid bridges (solid)
Absent bridges (dashed $\times$)

T2→T3
Not yet specified $\times$

T1→T2
Not yet specified $\times$

Double-bridge
calculation

1

(1) Bits processed per ignition
(~20 transmitted bits/event, bandwidth-derived)

2

(2) $\times$ Landauer minimum
($\approx 3 \times 10^{-21}$ J/bit at 310K; precisely $\approx 2.97 \times 10^{-21}$ J/bit)

3

(3) $\times$ neural-inefficiency factor $\approx 1.7 \times 10^{18} \times$
(the framework's primary Tier-1 falsification target)

4

(4) compare with PET/BOLD expenditure
(currently unvalidated)

T3→T2: Approximate-Inference Bridge
variational F minimisation $\rightarrow$ MI maximisation
Status: derived in ML, unconverged in vivo

T2→T1: Landauer bridge
 $E \geq kT \cdot \ln 2$ per bit erased
 $k \approx 100$ ATP/bit $\approx 5 \times 10^{-18}$ J/bit
Status: theoretically bounded, k unmeasured

T3→T1: Double bridge (3→2→1)
requires both above
Status: composite, unvalidated

Downward bridge (specified)
Upward bridge (deliberately unspecified)

Computational (Tier 3)
Information-theoretic (Tier 2)

Thermodynamic (Tier 1)
Empirical comparison

Footnote: phenomenal level L4 (Φ -boundary) is excluded by design (§8.4)